KHOA BUI

San Jose, CA | (669) 233-1475 | khoathanhbui@gmail.com | linkedin.com/in/khoa-bui | https://www.khoabui1993.github.io

Summary

Early-career engineer with experience in EDA systems, software testing, hardware validation, and IT infrastructure. Built Python and Bash regression automation, led hardware-in-the-loop testing for NXP Layerscape processors, and developed internal dashboards and provisioning tools. Brings hands-on experience across software, firmware, and laboratory systems to improve engineering workflows, system reliability, and testing efficiency.

TECHNICAL SKILLS

- **Systems & Virtualization**: Linux (RHEL, Ubuntu, Debian), Proxmox VE, KVM, PXE Boot, Ansible Automation
- **Hardware & Infrastructure**: Dell PowerEdge, HPE ProLiant, iDRAC, iLO, IPMI, RAID (PERC, LSI), NVMe/SAS, PDU, Cabling
- **Firmware & Test Validation**: CodeWarrior, POST Validation, BIOS/UEFI, Firmware Flashing, Hardware-in-the-Loop (HIL)
- **Scripting, DB & Tools**: Python, Bash, C/C++, Java, SQL, Git, Jira, REST APIs, MySQL, SQLite3, MongoDB, Full Stack Development

WORK EXPERIENCE

Hughes Network Systems **Jul 2023 - Dec 2025**
EDA Engineer *Germantown, MD*

- Administered enterprise ECAD/EDA software ecosystems, troubleshooting workflows, resolving system bugs, and conducting technical training for engineering teams.
- Engineered Python and Bash-driven regression test suites to validate RTL designs, integrating automated execution pipelines and streamlining defect triage workflows.
- Spearheaded board bring-up and hardware-in-the-loop (HIL) testing for NXP Layerscape LA12xx processors, validating SDR baseband firmware, POST sequences, and BIOS/UEFI parameters.
- Developed and maintained internal web portals and telemetry dashboards that improved hardware Bill of Materials (BOM) tracking and lab resource allocation.
- Administered OpenLDAP directory services, managing centralized user authentication and role-based access control (RBAC) for engineering teams across shared laboratory infrastructure.
- Architected automated bare-metal provisioning pipelines using PXE boot, Ansible playbooks, and Bash scripts, accelerating RHEL deployments across high-density server fleets.
- Diagnosed and resolved complex component-level hardware faults across Dell PowerEdge and HPE ProLiant platforms; leveraged iDRAC, iLO, and IPMI for out-of-band management.
- Executed enterprise rack-and-stack deployments in hardware labs, delivering deployment-ready infrastructure through structured server cabling, balanced PDU power, accurate port labeling, and top-of-rack network uplink patching.

Vibrant America **Apr 2023 - Jul 2023**
Lab IT Support Specialist *Santa Clara, CA*

- Troubleshot full-stack laboratory diagnostic applications using Java, Python, and SQL, resolving service-logic issues to improve application reliability and performance.
- Designed relational database schemas and automated data verification scripts, improving data accuracy and strengthening clinical application reliability.
- Partnered with the Lab Test Development team to define test automation requirements and maintain system runbooks, improving testing efficiency and operational consistency.

Netgear Inc. **Dec 2021 - May 2022**
Software Quality Assurance Intern *San Jose, CA*

- Built a Python test automation tool utilizing PyQt and REST APIs, cutting regression execution cycles by 30%.
- Authored 100+ test cases and documented 40+ verified defects for Android networking applications using Jira.
- Validated production router firmware builds through manual and automated testing, using Firebase runtime telemetry to identify issues and ensure stable, release-ready releases.

Ezvps **Jan 2021 - Dec 2021**
IT / Systems Support Engineer *Ho Chi Minh City, Vietnam*

- Deployed and maintained multi-tenant virtualization clusters using Proxmox VE, KVM, and Linux containers, enabling reliable, scalable infrastructure for multiple customer environments.
- Configured and optimized RAID 0, 1, 5, and 10 arrays on PERC and LSI controllers, enabling high-performance, fault-tolerant storage pools for enterprise SAS, SATA, and NVMe workloads.

- Validated server stability and performance through stress testing with MemTest86, Prime95, and vendor diagnostics, identifying hardware issues before commissioning and ensuring reliable deployment.
- Delivered data center infrastructure by installing racks, deploying structured cabling, balancing PDU loads, and configuring switch uplinks to enable reliable power distribution and network connectivity.

EDUCATION

San Jose State University **2020 - 2022**
Bachelor of Science, Computer Science *San Jose, CA*
- **Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, NoSQL Systems, Relational Database

De Anza College **2018 - 2020**
Associate of Science, Computer Science *Cupertino, CA*